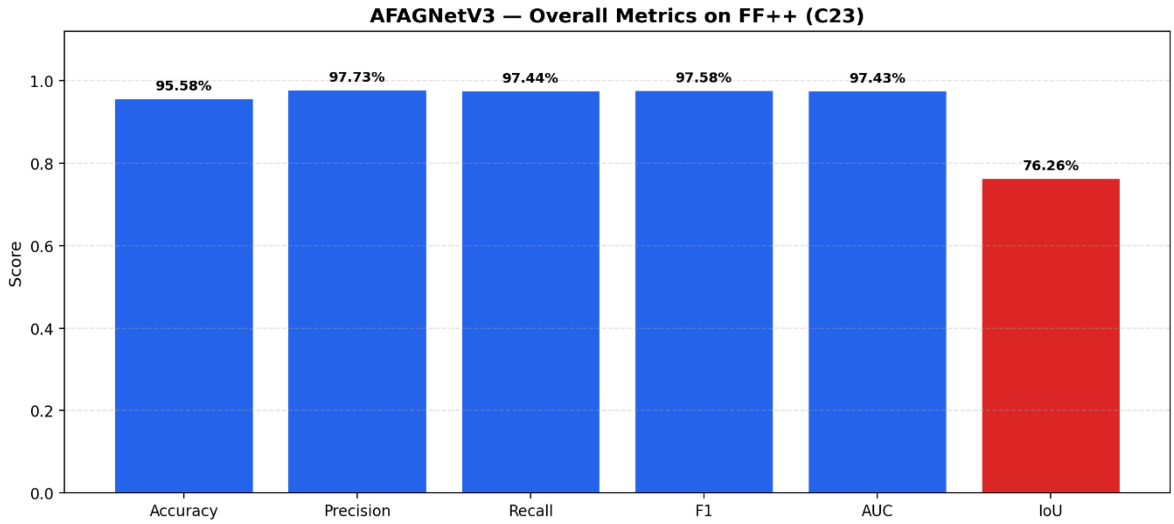



```
Accuracy      : 95.58%          F1          : 97.58%
Precision     : 97.73%          AUC         : 97.43%
Recall        : 97.44%          IoU         : 76.26%
```

1. Accuracy

Accuracy shows how many predictions the model got right out of all the predictions. It gives idea of overall performance but it can be misleading when one class is more dominant over the other. For example a model that predicts the majority class correctly most of the time might have high accuracy but still fail to capture important details about other classes. It can be calculated using the below formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2. Precision

Precision focus on the quality of the model's positive predictions. It tells us how many of the "positive" predictions were actually correct. It is important in situations where false positives need to be minimized such as detecting spam emails or fraud. The formula of precision is:

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Recall

Recall measures how how good the model is at predicting positives. It shows the proportion of true positives detected out of all the actual positive instances. High recall is essential when missing positive cases has significant consequences like in medical tests.

$$\text{Recall} = \frac{TP}{TP + FN}$$

4. F1-Score

F1-score combines precision and recall into a single metric to balance their trade-off. It provides a better sense of a model's overall performance particularly for imbalanced datasets. It is helpful when both false positives and false negatives are important though it assumes precision and recall are equally important but in some situations one might matter more than the other.

$$\text{F1-Score} = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall}$$

6. Type 1 and Type 2 error

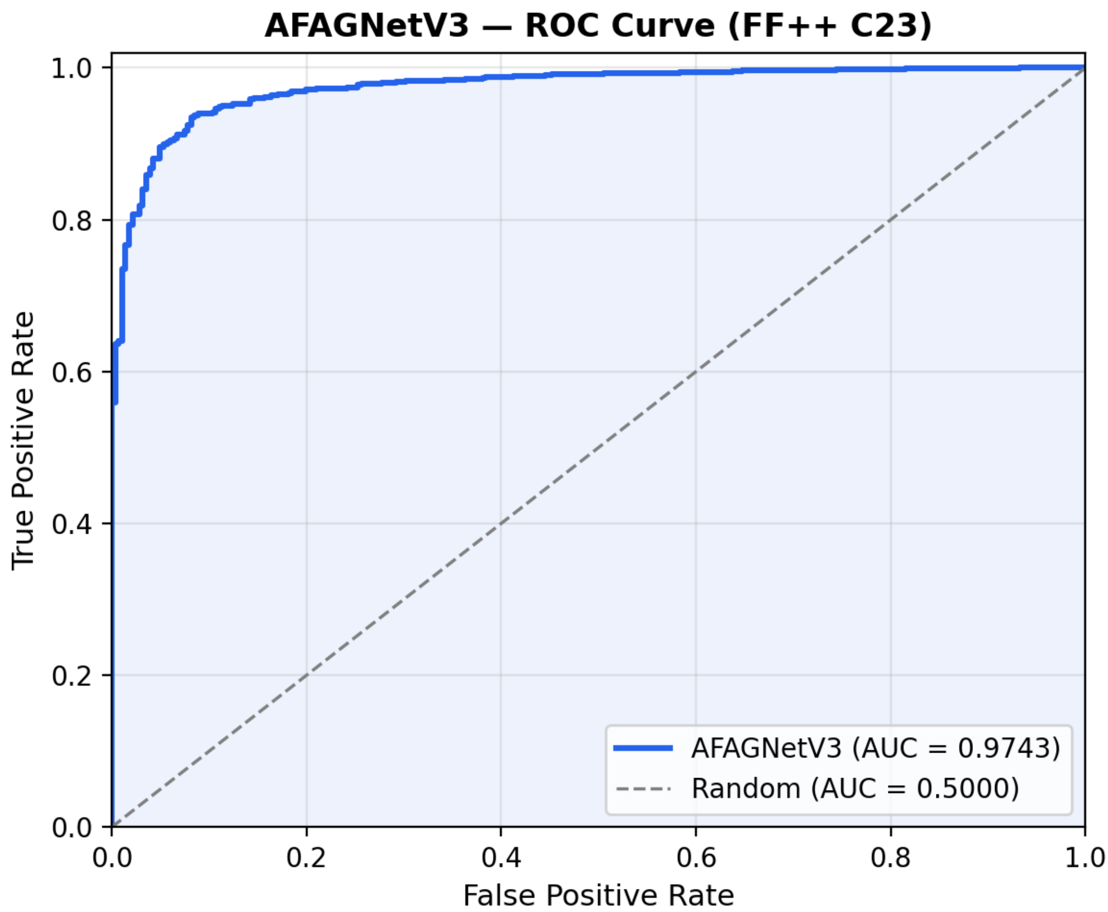
Type 1 and Type 2 error are:

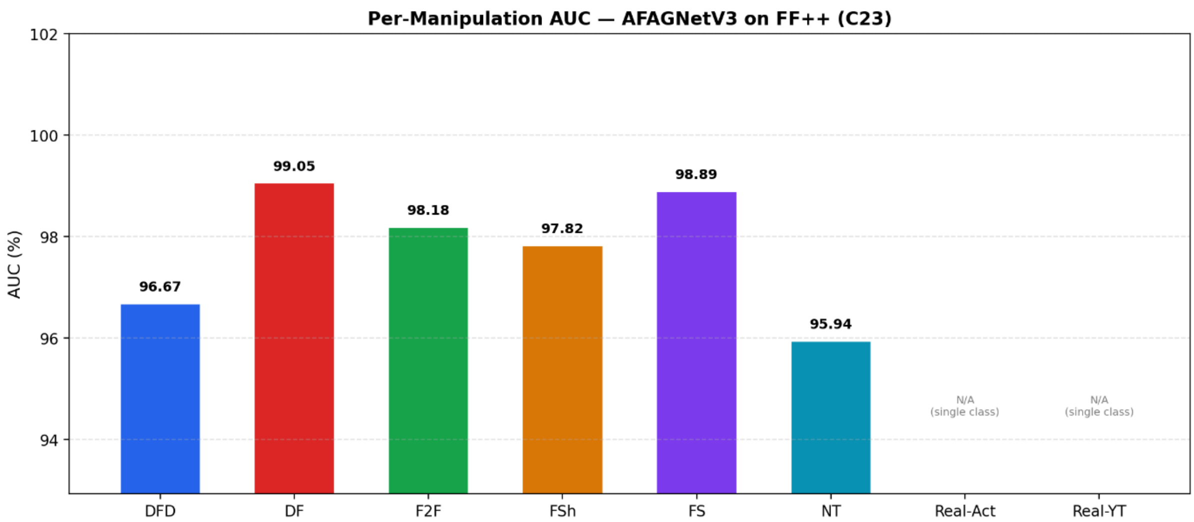
- **Type 1 error**: It occurs when the model incorrectly predicts a positive instance but the actual instance is negative. This is also known as a **false positive**. Type 1 Errors affect the **precision** of a model which measures the accuracy of positive predictions.

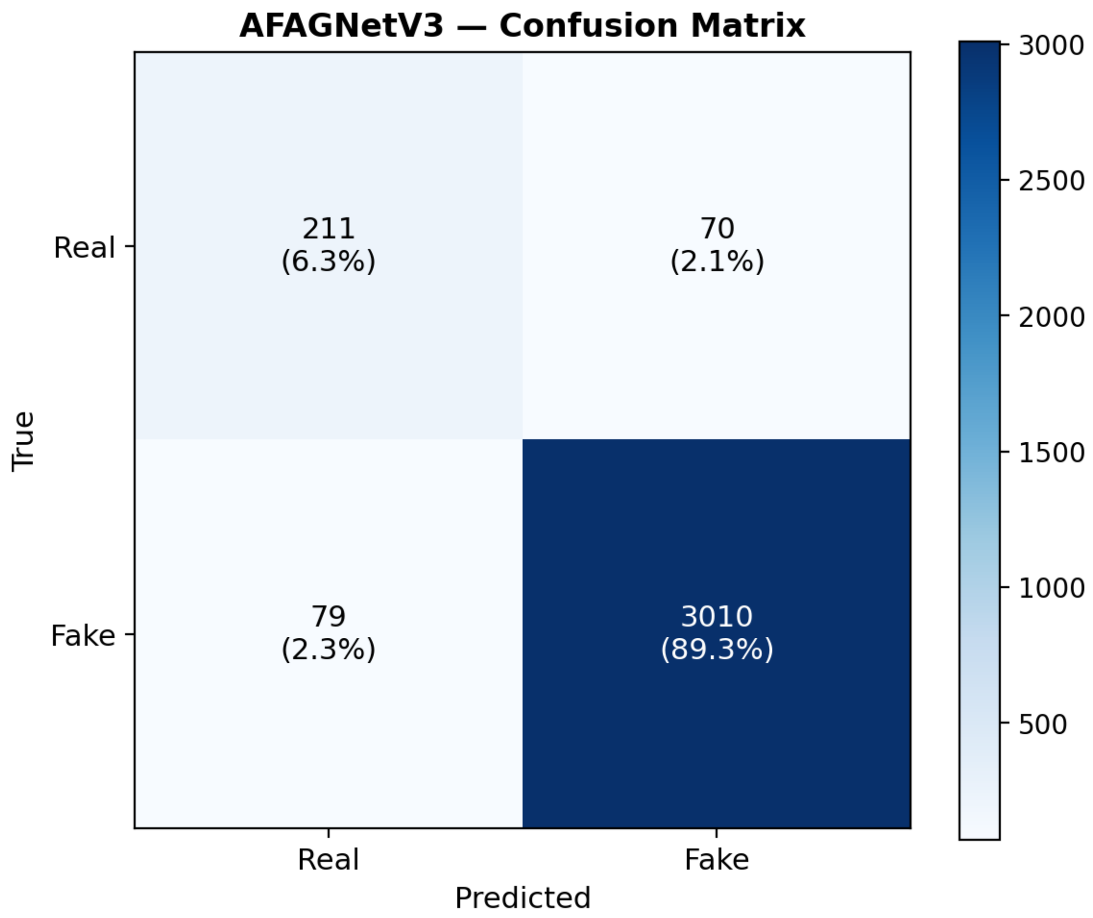
$$\text{Type 1 Error} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

- **Type 2 error**: This occurs when the model fails to predict a positive instance even though it is actually positive. This is also known as a **false negative**. Type 2 Errors impact the **recall** of a model which measures how well the model identifies all actual positive cases.

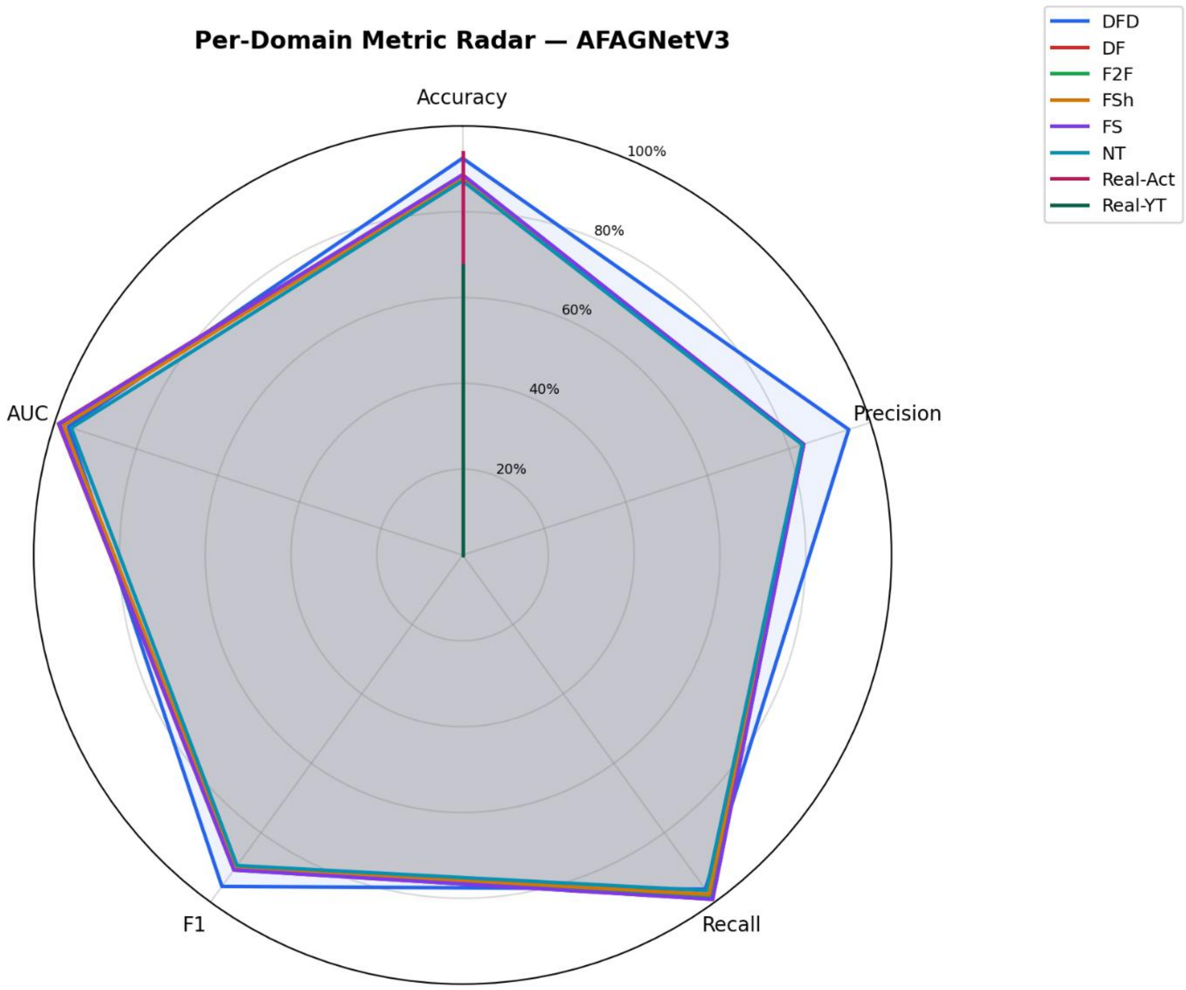
$$\text{Type 2 Error} = \frac{FN}{TP + FN}$$

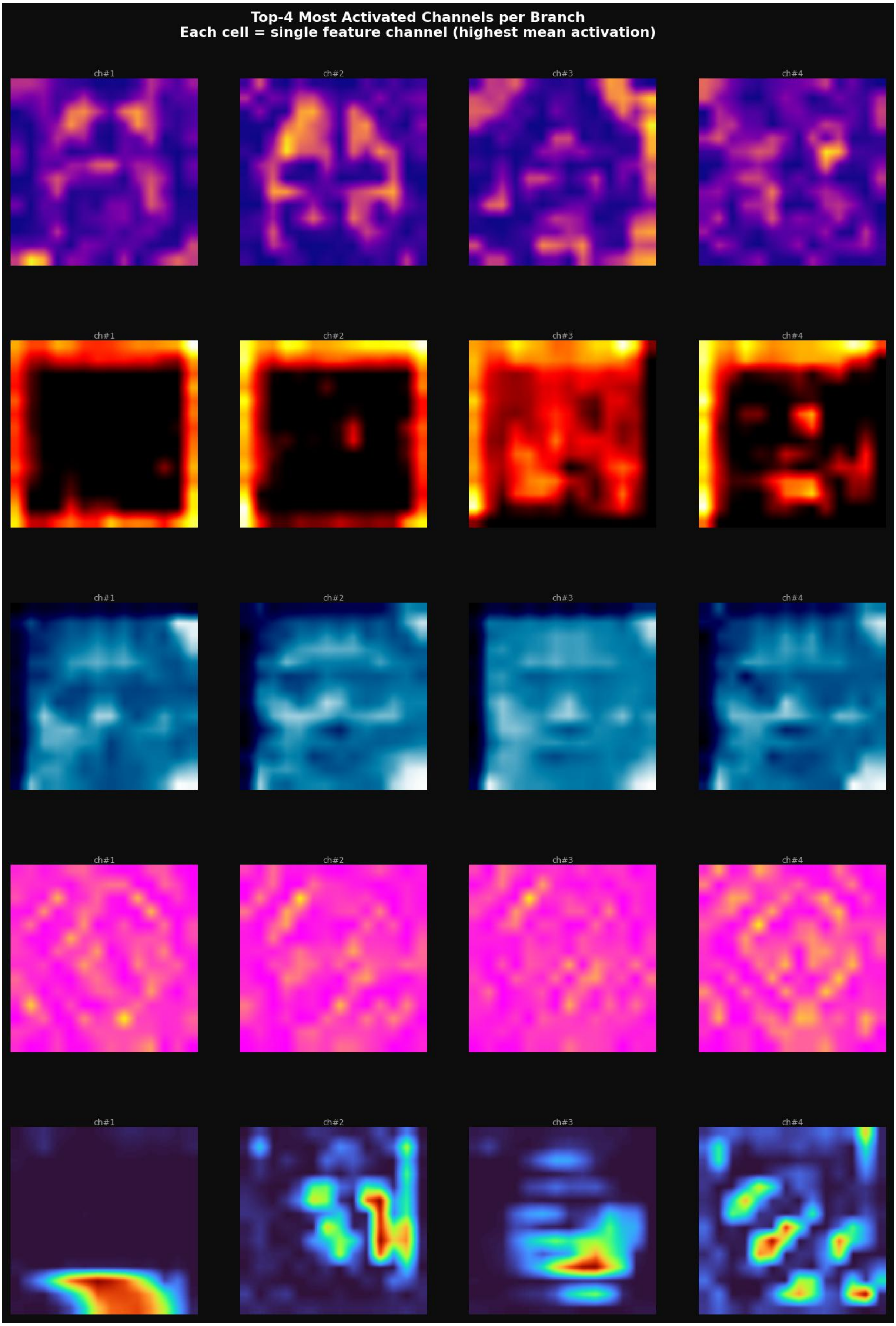

AFAGNetV3 — ROC Curve (FF++ C23)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
AFAGNetV3 (AUC = 0.9743)
Random (AUC = 0.5000)

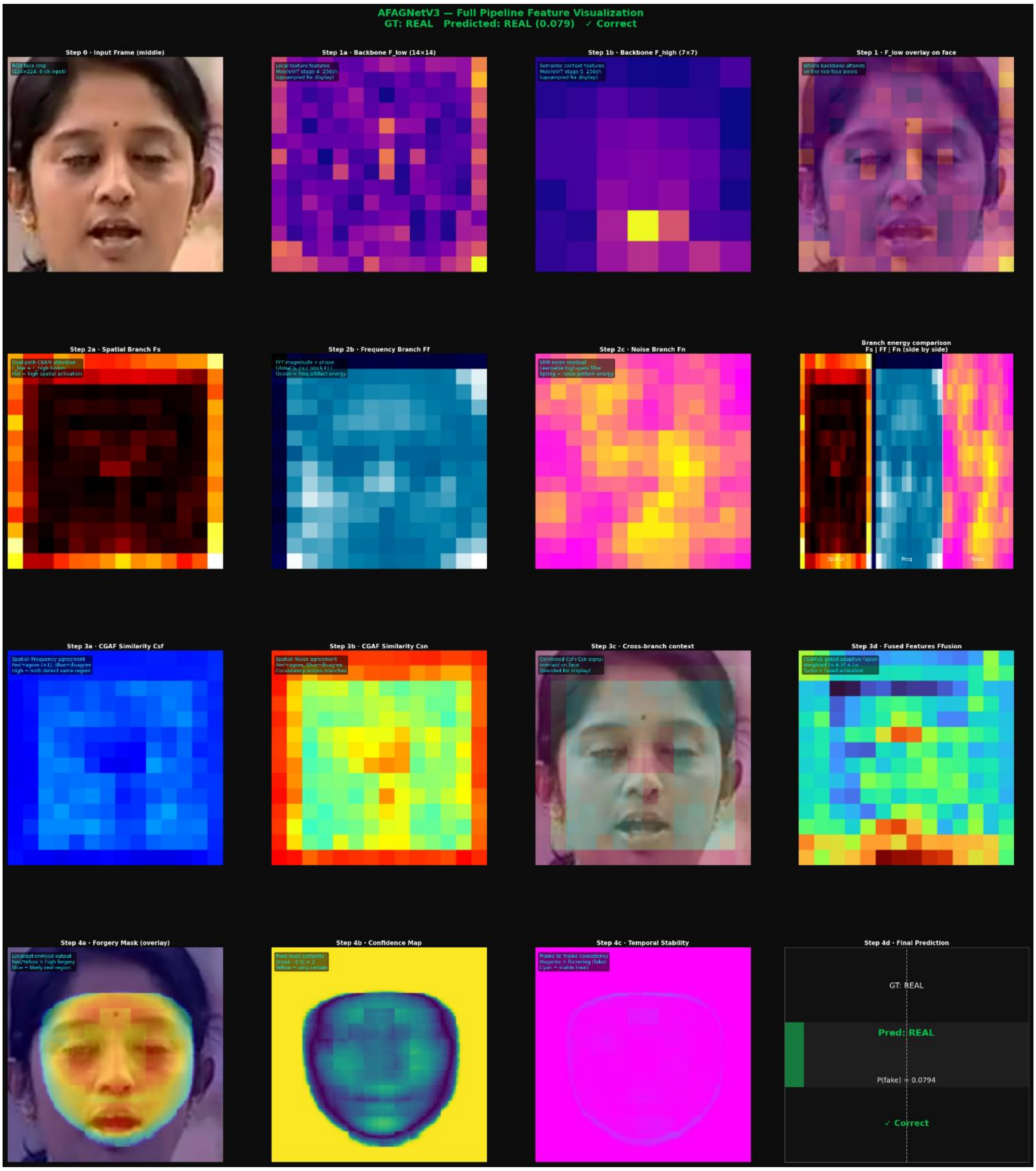

Per-Manipulation AUC — AFAGNetV3 on FF++ (C23)
AUC (%)
102
100
98
96
94
96.67
99.05
98.18
97.82
98.89
95.94
N/A
(single class)
N/A
(single class)
DFD
DF
F2F
FSh
FS
NT
Real-Act
Real-YT

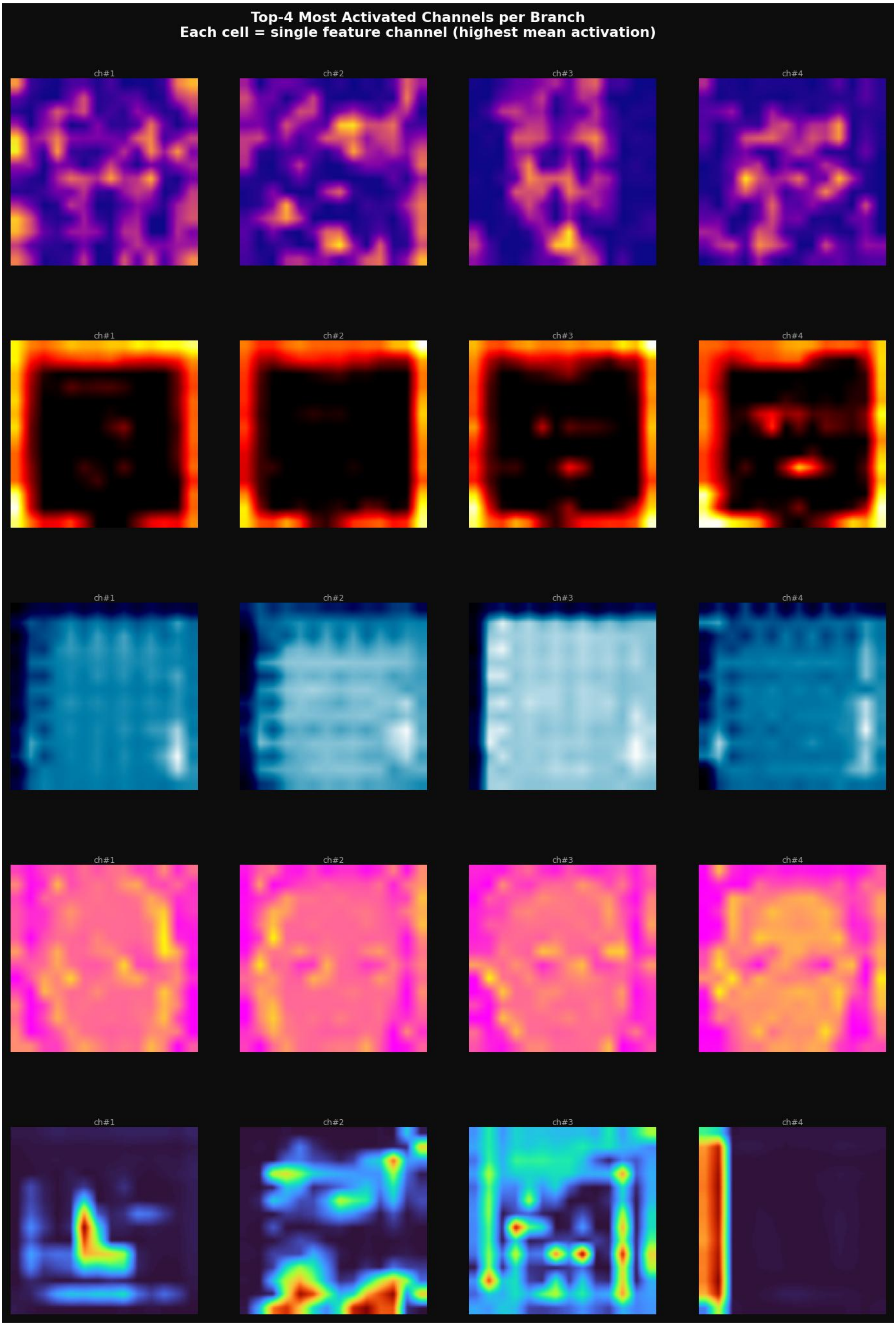


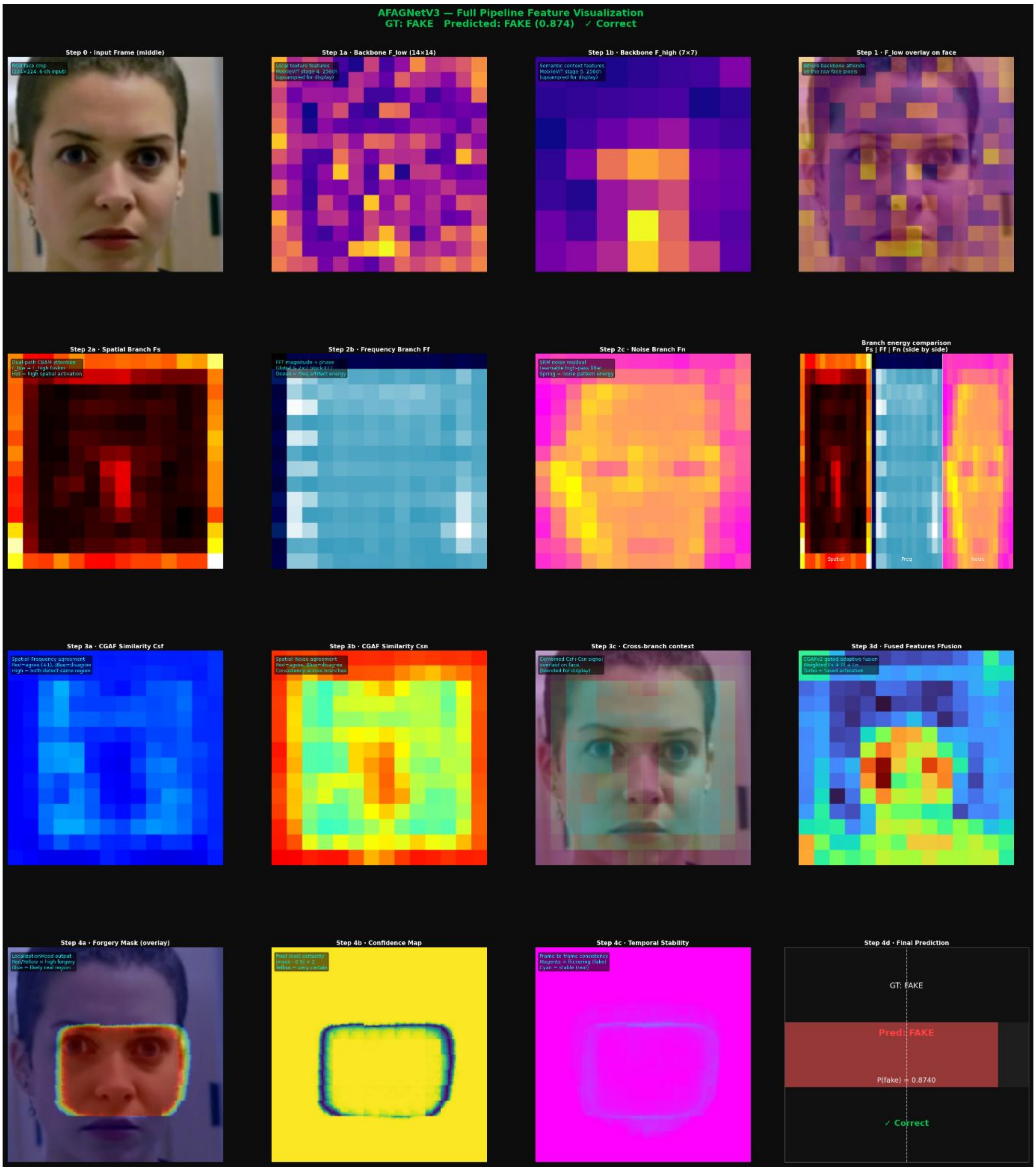
AFAGNetV3 — Confusion Matrix
211
(6.3%)
70
(2.1%)
79
(2.3%)
3010
(89.3%)
Real
Fake
True
Real
Fake
Predicted
3000
2500
2000
1500
1000
500


Per-Domain Metric Radar — AFAGNetV3
Accuracy
Precision
Recall
F1
AUC
20%
40%
60%
80%
100%
DFD
DF
F2F
FSh
FS
NT
Real-Act
Real-YT


Top-4 Most Activated Channels per Branch
Each cell = single feature channel (highest mean activation)
ch#1
ch#2
ch#3
ch#4
ch#1
ch#2
ch#3
ch#4
ch#1
ch#2
ch#3
ch#4
ch#1
ch#2
ch#3
ch#4
ch#1
ch#2
ch#3
ch#4


AFAGNetV3 — Full Pipeline Feature Visualization
GT: REAL Predicted: REAL (0.079) ✓ Correct
Step 0 · Input Frame (middle)
Step 1a · Backbone F_low (14×14)
Step 1b · Backbone F_high (7×7)
Step 1 · F_low overlay on face
Step 2a · Spatial Branch Fs
Step 2b · Frequency Branch Ff
Step 2c · Noise Branch Fn
Branch energy comparison
Fs | Ff | Fn (side by side)
Step 3a · CGAF Similarity Csf
Step 3b · CGAF Similarity Csn
Step 3c · Cross-branch context
Step 3d · Fused Features Ffusion
Step 4a · Forgery Mask (overlay)
Step 4b · Confidence Map
Step 4c · Temporal Stability
Step 4d · Final Prediction
GT: REAL
Pred: REAL
P(fake) = 0.0794
✓ Correct

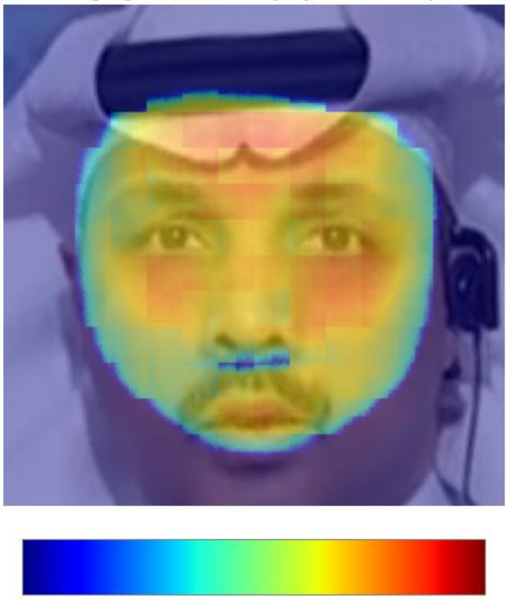

Top-4 Most Activated Channels per Branch
Each cell = single feature channel (highest mean activation)
ch#1
ch#2
ch#3
ch#4


AFAGNetV3 — Full Pipeline Feature Visualization
GT: FAKE Predicted: FAKE (0.874) ✓ Correct
Step 0 · Input Frame (middle)
Step 1a · Backbone F_low (14×14)
Local texture features
MobileViT stage 4, 256ch
(upsampled for display)
Step 1b · Backbone F_high (7×7)
Semantic context features
MobileViT stage 5, 256ch
(upsampled for display)
Step 1 · F_low overlay on face
Where backbone attends
on the raw face pixels
Step 2a · Spatial Branch Fs
Dual-path CBAM attention
Hot = high spatial activation
Step 2b · Frequency Branch Ff
FFT magnitude + phase
Step 2c · Noise Branch Fn
SRM noise residual
Learnable high-pass filter
Spring = noise pattern energy
Branch energy comparison
Fs | Ff | Fn (side by side)
Spatial
Freq
Noise
Step 3a · CGAF Similarity Csf
Spatial-Frequency agreement
High = both detect same region
Step 3b · CGAF Similarity Csn
Spatial-Noise agreement
Red=agree, Blue=disagree
Consistency across branches
Step 3c · Cross-branch context
overlaid on face
(blended for display)
Step 3d · Fused Features Ffusion
CGAFv2 gated adaptive fusion
Turbo = fused activation
Step 4a · Forgery Mask (overlay)
LocalizationHead output
Red/Yellow = high forgery
Blue = likely real region
Step 4b · Confidence Map
Pixel-level certainty
Yellow = very certain
Step 4c · Temporal Stability
Frame to frame consistency
Magenta = flickering (fake)
Cyan = stable (real)
Step 4d · Final Prediction
GT: FAKE
Pred: FAKE
P(fake) = 0.8740
✓ Correct

Confidence Map [VIRIDIS colormap]

Yellow = model is certain | Dark purple = uncertain / low activation
Yellow background = outside face mask (ignored region).

Forgery Mask Overlay [JET colormap]



Red/Yellow = high forgery probability | Blue = low probability (likely real)
Real faces → diffuse attention. Fake faces → tight rectangle = blending boundary.

Temporal Stability [COOL colormap]

Cyan = consistent across frames (real indicator) | Magenta = flickering (fake indicator)
Solid magenta background = zero gradient / model is very certain.